

APOLLO AGRIVERSE

FastAPI Backend Integration & Validation Report

Professional Technical Documentation

Project	Apollo AgriVerse
System area	Backend API and integrated farm suitability evaluation
Framework	FastAPI
Server	Uvicorn / ASGI
Primary endpoint	POST /api/evaluate
Validation	HTTP 200 — successful evaluation request
Scope	Core suitability evaluation API

This report documents the completed FastAPI backend phase, the integration path used by the application, the request/response contract, and the validation results obtained during local testing.

1. Executive Summary

The FastAPI layer was implemented to expose the existing Apollo AgriVerse farm-suitability workflow through HTTP. The implementation connects the existing knowledge base, external-data service, ML engine service and integrated suitability engine without rebuilding the underlying agronomic or machine-learning components.

The backend was started with Uvicorn, the knowledge-base tables were loaded successfully, the existing PKL-based ML assets were initialized, the health endpoint responded successfully, and the main farm-evaluation endpoint returned HTTP 200 with a successful evaluation report.

2. Backend Architecture



The FastAPI layer is therefore an API boundary around the existing evaluation logic. Its responsibility is to receive validated farm input, invoke the integrated backend service, and return the resulting evaluation as structured JSON.

3. Integrated Backend Components

Component	Responsibility
KnowledgeBaseLoader	Loads and validates the crop, variety, soil, crop-soil requirement and Maharashtra region/climate knowledge base.
ExternalDataService	Provides external information used by the evaluation workflow, including configured weather/market data.
MLEngineService	Initializes the existing PKL models and scalers used by the backend ML workflow.
connectSuitabilityEngine	Provides the integrated suitability-evaluation entry point and receives the existing backend services.
FastAPI application	Defines the HTTP API, validates requests and returns JSON responses.
Uvicorn	Runs the FastAPI application as the local ASGI server.

4. API Specification

Method	Endpoint	Purpose	Status
GET	/health	Backend health and integrated configuration check.	PASS
POST	/api/evaluate	Farm suitability evaluation and recommendation response	PASS

No additional endpoints were introduced during this phase. The API surface was intentionally kept limited to the core evaluation workflow already defined for the project.

5. Farm Evaluation Request

```
POST /api/evaluate
Content-Type: application/json

{
  "farm_id": "FARM_MH_NASHIK_01",
  "region_id": "REG_MH_NASHIK_01",
  "soil_id": "SOIL_BLACK_MH_01",
  "water_availability": "medium",
  "latitude": 20.0059,
  "longitude": 73.7897
}
```

The request was submitted through the FastAPI Swagger interface. This validates the complete HTTP path rather than only checking whether the Python modules can be imported.

6. Validation Results

Validation item	Result	Interpretation
FastAPI/Uvicorn startup	PASS	Application server started successfully.
Knowledge Base initialization	PASS	Maharashtra knowledge-base tables loaded.
ML Engine initialization	PASS	Existing PKL models/scalers were reported as loaded.
Integrated engine initialization	PASS	Integrated suitability path initialized.
GET /health	PASS	Health/configuration response received.
POST /api/evaluate	PASS	Actual evaluation request returned HTTP 200.
Response success flag	PASS	Response contained success=true.

7. Evaluation Response — Observed Outputs

Output category	Observed value
Location	Nashik, Maharashtra
Soil profile	Black Soil
Focus crop	GRAPE
Final suitability score	86.2

Output category	Observed value
Agronomic score	88.8
Expected yield	12.32 tons/ha
Suitability band	Highly Suitable
Climate score	100.0
Soil score	82.6
Water score	83.3
Market score	75.0
Market price	■120/kg

The response also included evaluation pros/cons and alternative crop recommendations.

8. Current Implementation Status

Area	Status
Knowledge Base	Completed / operational
Suitability Engine integration	Completed / operational
ML Engine / PKL integration	Completed / operational
FastAPI application	Completed / operational
Health endpoint	Verified
Farm evaluation endpoint	Verified with HTTP 200
Frontend integration	Separate next phase

9. Scope Boundaries

- The FastAPI phase does not retrain or modify the underlying PKL models.
- The FastAPI phase does not replace the existing suitability-engine logic.
- The FastAPI phase does not require rebuilding the existing frontend.
- Additional endpoints should only be added if they are required by the agreed project workflow.
- A successful HTTP response confirms API execution and integration; it does not by itself establish scientific correctness of every underlying scoring rule.

10. Recommended Next Phase

With the core FastAPI evaluation layer operational, the next phase can focus on the existing frontend and how it will consume the API response. The backend API should remain stable while the frontend request fields and response mapping are reviewed.

This keeps the project controlled: first establish a stable backend contract, then connect the already-developed frontend to that contract, and finally perform end-to-end testing.

11. Conclusion

The Apollo AgriVerse FastAPI backend has reached a functional milestone. The application starts successfully, the integrated backend services initialize, the health endpoint responds, and a real farm evaluation request successfully returns a structured JSON result with HTTP 200.

FASTAPI CORE VALIDATION STATUS: SUCCESSFULLY OPERATIONAL